

Cameron Lewis

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EDUCATION

Memorial University of Newfoundland

Bachelor of Engineering (Co-op), Electrical Engineering; GPA: 3.7/4.0

St. John's, NL, Canada

2024 – 2029

EXPERIENCE

Nokia

HW/SW Integration Engineering Co-op, NI IP Hardware Development

Ottawa, ON, Canada

Sept. – Dec. 2026

- Developed Python test automation scripts for optical drive testing and validation, replacing manual measurement sequences with repeatable characterization runs.
- Recorded wavelength, SMSR, and other optical transmitter and receiver characteristics on an Optical Spectrum Analyzer, verifying each against datasheet specifications to validate drive performance for the Hardware Optical and Hardware Timing teams.

Kraken Robotics Systems Inc.

Electrical Engineering Co-op

Mount Pearl, NL, Canada

Jan. – Apr. 2026

- Designed and revised PCBs in Altium for subsea platforms including Synthetic Aperture Sonar and the Katfish ROTV, carrying boards from schematic capture through layout review across multiple product lines.
- Redesigned wiring harness layouts against signal integrity review feedback from senior engineers, reducing failure points across cable assemblies.
- Bench-validated circuit designs with oscilloscopes against performance and manufacturability specifications.
- Maintained technical documentation for 5+ products in PDM software, enabling accurate revision control and streamlining design review approvals.

Paradigm Engineering

Electrical Team Lead

St. John's, NL, Canada

Sept. 2025 – Present

- Designed multilayer control PCBs in KiCad integrating microcontrollers, power-management circuits, and voltage regulators for the autonomous kart campaigned at the 2026 Autonomous Karting Series at Purdue University.
- Architected the kart's 4-layer Auxiliary Control Board end to end as lead control module designer, specifying an STM32F4 microcontroller, CAN and UART links, a 5 V to 3.3 V LDO logic rail, and level shifting and protection circuitry feeding every electrical subsystem.
- Brought up and debugged assembled boards with oscilloscopes and multimeters, validating voltage rails and signal behaviour against design specifications before vehicle integration.
- Led a team of 10+ members across board design, assembly, and bring-up of the kart's electrical subsystems.

PROJECTS

Auxiliary Control Board (Paradigm)

Apr. 2026

- Designed a 4-layer auxiliary control PCB in KiCad around an STM32F4 microcontroller, integrating CAN and UART interfaces, a USB-to-UART bridge for debug and flashing, a 5 V to 3.3 V LDO logic rail, and level-shifting and protection circuitry for distribution across kart subsystems.
- Qualified the assembled board through thermal and functional testing, confirming rail stability and signal integrity before it was released to the vehicle.

Brick Buck Board (Paradigm)

Apr. – May 2026

- Designed a 4-layer power distribution PCB integrating two 48 V to 12 V bucks and a 12 V to 5 V buck stage, fronted by a hot-swap battery input protection circuit.
- Supplied regulated power to the kart's cooling, braking, steering, e-stop, Jetson AGX Orin, and control board subsystems, qualifying regulation and thermal stability under load across the full operating range.

SKILLS

Hardware	Digital & Analog Circuit Design, Multilayer PCB Design, Power Electronics, DC-DC Converter Design, Signal Integrity
PCBA & Test	SMD & Through-Hole Soldering, Solder Paste & Reflow Oven Assembly, Rework, Board Bring-Up, Oscilloscopes, Logic Analyzers, Multimeters
Protocols	CAN, UART, USB, I2C, SPI
Languages	C, C++, Python, MATLAB, Bash, RISC-V Assembly
Tools	Altium Designer, KiCad, LTspice, PSpice, Cadence Allegro, Git, PDM, Colab, SR Linux, SR OS, pySROS, Pyshl